

An international study on THM modelling of the full-scale heater experiment at Mont Terri laboratory

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ABSTRACT

We present results from an international model comparison study of the Full-Scale Emplacement (FE) experiment in Opalinus Clay at the Mont Terri Laboratory, Switzerland. Based on a provided parameter set the teams decided which parameters they adopted for their models, whether they considered the excavation and the ventilation phase in addition to the heating phase and if they included technical features like the shotcrete or the EDZ. The teams were able to reproduce the measured parameters temperature, relative humidity and pore pressure. The modelled results for temperature agree very closely between the teams especially in the sensors in Opalinus Clay. All teams were able to reproduce the redistribution of water in the bentonite backfill due to heating. The evolution of the relative humidity showed similar trends with differences in the intensity of the dry out effect. To model the pore pressure evolution is more complex because it comprises the full interaction of the coupled THM processes. The spread between the pore pressure modelled by the teams was larger, with some teams overestimating the pressure increase due to heating and some teams overestimating the extent of drainage. The agreement of modelled results with measurements improves with larger distance to the heater. We conclude that the EDZ and the shotcrete potentially influence the behaviour of the rock causing higher differences closer to the heater. Further research is needed to better implement those influences into the models. Based on the calibrated models, the future evolution of temperature, relative humidity and pore pressure was predicted over the next 10 years following a change of the heat power applied in 2023 and 2024. Again, the predicted temperatures agree very closely between the teams. Most teams do not expect an increase in relative humidity during the next 10 years after the initial dry-out.

Introduction

Numerical models are used to assess the long-term performance of

geological disposal systems for radioactive waste. To build confidence in these models, it is necessary to test the models against data from lab scale to large scale experiments. One example is the Full-Scale

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Emplacement (FE) experiment at the Mont Terri rock laboratory in Switzerland. It is a 1:1 scale representation of the Swiss repository concept, and one of the motivations for this experiment was to provide data for validation of existing numerical models of coupled thermo-hydro-mechanical (THM) processes within the Opalinus Clay host rock, at the full scale for geological disposal.²⁴

The objective of this study is to model the coupled thermal-hydraulic-mechanical processes of the FE experiment with focus on the pore pressure evolution in Opalinus Clay. This focus will enable the teams to reproduce the measured pore water pressure and to predict the future evolution due to changes in the course of the experiment. This paper presents the work of nine modelling teams (Table 1) who undertook the modelling as part of the recently completed phase of DECOVALEX,⁴ DECOVALEX-2023 (D-2023).

BGE Bundesgesellschaft für Endlagerung mbH, TUBAF Technische Universität Bergakademie Freiberg, BGR Federal Institute for Geosciences and Natural Resources, UFZ Helmholtz Centre for Environmental Research, CAS State Key Laboratory of Geomechanics and Geotechnical Engineering, Institute of Rock and Soil Mechanics, Chinese Academy of Sciences, DOE US Department of Energy, LBNL Lawrence Berkeley National Laboratory, ENSI Swiss Federal Nuclear Safety Inspectorate, GRS Gesellschaft für Anlagen- und Reaktorsicherheit (GRS) gGmbH, Braunschweig, Germany, KAERI Korea Atomic Research Institute, KIGAM Korea Institute of Geoscience and Mineral Resources, NWMO Nuclear Waste Management Organization, DOE US Department of Energy/ SNL Sandia National Laboratories.

Teams participated in a staged modelling programme of gradually increasing complexity, culminating in a 3D-calibrated representation of the FE experiment. The steps that were taken are:

- Step 0: 2D benchmarking of models, with prescribed geometry, processes, boundary conditions and parameters.
- Step 1a: 3D benchmarking of models, with prescribed geometry, processes, boundary conditions and parameters (not presented in this paper).
- Step 1c: parameter calibration of the 3D model to data (not presented in this paper).
- Step 2: 3D model of the FE experiment including ventilation and any additional features.
- Step 3: Extended heating prediction.

Modelling of similar experiments in argillaceous claystone has been undertaken in previous phases of DECOVALEX (¹² & b,³⁸), giving a strong foundation for both the processes that needed to be included in the models and the parameter values used to describe the Opalinus Clay. Benchmarking steps were designed so that all teams used this previous knowledge and started from the same assumptions about important processes and parameters. Once teams obtained similar results in the 2D and 3D models, they calibrated their models against the data, including

both parameter calibration and addition of features and processes to the model.

The work undertaken by the teams revealed the benefits of both taking a stepwise approach to building complex models and working with multiple independent teams, providing a cross-check on each other's work. Most teams were able to represent the main features of the data, giving confidence that the broad understanding of THM processes and properties of the Opalinus Clay is good. Some details in the hydro-mechanical behaviour of the Opalinus Clay close to the tunnel were captured less well by some teams, indicating the need for better understanding of processes and properties where the Opalinus Clay may have been affected by excavation.

Mont Terri FE experiment

The Full-Scale Emplacement (FE) experiment at the Mont Terri rock laboratory in Switzerland is designed to simulate a spent fuel / vitrified high-level waste disposal tunnel comparable with the Swiss repository concept.²⁴ The disposal concept consists of 4.6 m long cylindrical waste canisters lying horizontally along tunnels of several hundred meters within Opalinus Clay.²⁵ The waste canisters lie on pedestals made from compacted bentonite blocks and a granular bentonite backfill is emplaced around the canisters to fill the remaining space in the tunnels.²⁴ At that time the tunnel lining was foreseen as shotcrete.

The FE experiment is in the Opalinus Clay formation, the envisaged host rock for the Swiss repository concept. The Opalinus Clay at the Mont Terri laboratory is an over-consolidated bedded claystone and exhibits a strong anisotropy of the key material properties like thermal conductivity, permeability and Young's modulus. At Mont Terri, the Opalinus Clay has undergone tectonic deformation and is locally folded, which leads to a bedding angle of 34°. The potential sites for the Swiss repository are foreseen in Northern Switzerland, where the Opalinus Clay has not undergone tectonic deformation and the bedding is close to horizontal.

The FE experiment consists of a 50 m long tunnel through Opalinus Clay with three heaters of dimensions similar to waste canisters emplaced on top of bentonite block pedestals (Fig. 1). The remaining space was filled by granular bentonite emplaced as densely and homogeneously as possible with a prototype backfilling machine with 5 screw conveyors.²⁴

The FE experiment is highly instrumented²⁴ with boreholes drilled into the Opalinus Clay to monitor temperature, pressure and deformation (Fig. 2). Within the tunnel, sensors were mounted to the tunnel wall, the heaters and the bentonite blocks to measure parameters including temperature, relative humidity and deformation. Several hundred sensors were installed giving a large database of information against which models can be compared.

The FE tunnel was excavated parallel to bedding after installing the sensors in the monitoring boreholes. A road header was used for excavation, and it took from April 2012 until July 2012 to excavate the 50 m long tunnel with a nominal diameter of 3 m. Excavation steps were limited to 1 m to 1.5 m per day. The tunnel support consists of rock bolts, mesh-reinforced shotcrete and steel arches.²⁶ With the support the tunnel has an inner diameter of 2.48 – 2.68 m and an outer diameter of 3.0 – 3.06 m. The excavation was followed by a ventilation period from May 2012 until February 2015 which was observed by sensors for temperature and relative humidity within the tunnel. At the end of 2014, the heater elements were emplaced in the tunnel, the tunnel was back-filled, and the heaters were switched on in sequence (Table 2). From February to March 2015 the concrete plug was installed. The heat power of 1350 W corresponds to the expected initial heat power of the waste canisters.²⁶ Initially a heat power of 1500 W was planned but the resulting temperatures were expected to be close to the maximum working limit of different sensors. There is no detailed information available on the extent of the Excavation Damaged Zone (EDZ) that was formed during the excavation.

Table 1

Teams participating in modelling the FE-Experiment in DECOVALEX-2023 and the codes that they used.

Team	Code	Citations for Code
BGE/ TUBAF	OpenGeoSys 6	Bilke et al., ² Lehmann et al. ²³
BGR/UFZ	OpenGeoSys 5 and 6	Kolditz et al. (2014), Bilke et al., ² Grunwald et al., ¹⁵
CAS	CASRock	Hou et al. ¹⁹
DOE/LBNL	TOUGH-FLAC	Rutqvist, ^{34,35} Rinaldi et al., ³³
ENSI	OpenGeoSys 6	Bilke et al., ² Lehmann et al. ²³
GRS	CODE BRIGHT	Olivella et al. ²⁹
KAERI/ KIGAM	OpenGeoSys 5 and OpenGeoSys-FLAC3D	Kolditz et al. (2014), Kim et al. ²¹
NWMO	COMSOL Multiphysics ®	COMSOL ⁹
DOE/SNL	PFLOTRAN, COMSOL Multiphysics ®	Hammond et al., ¹⁸ COMSOL ⁹

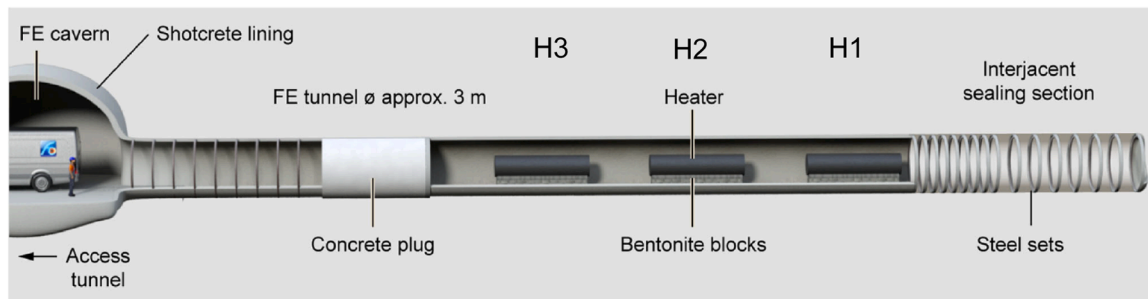


Fig. 1. Diagram showing the layout of the FE experiment at the Mont Terri rock laboratory (modified from²⁴). The granular bentonite surrounding the heaters is not shown within the 50 m long FE tunnel.

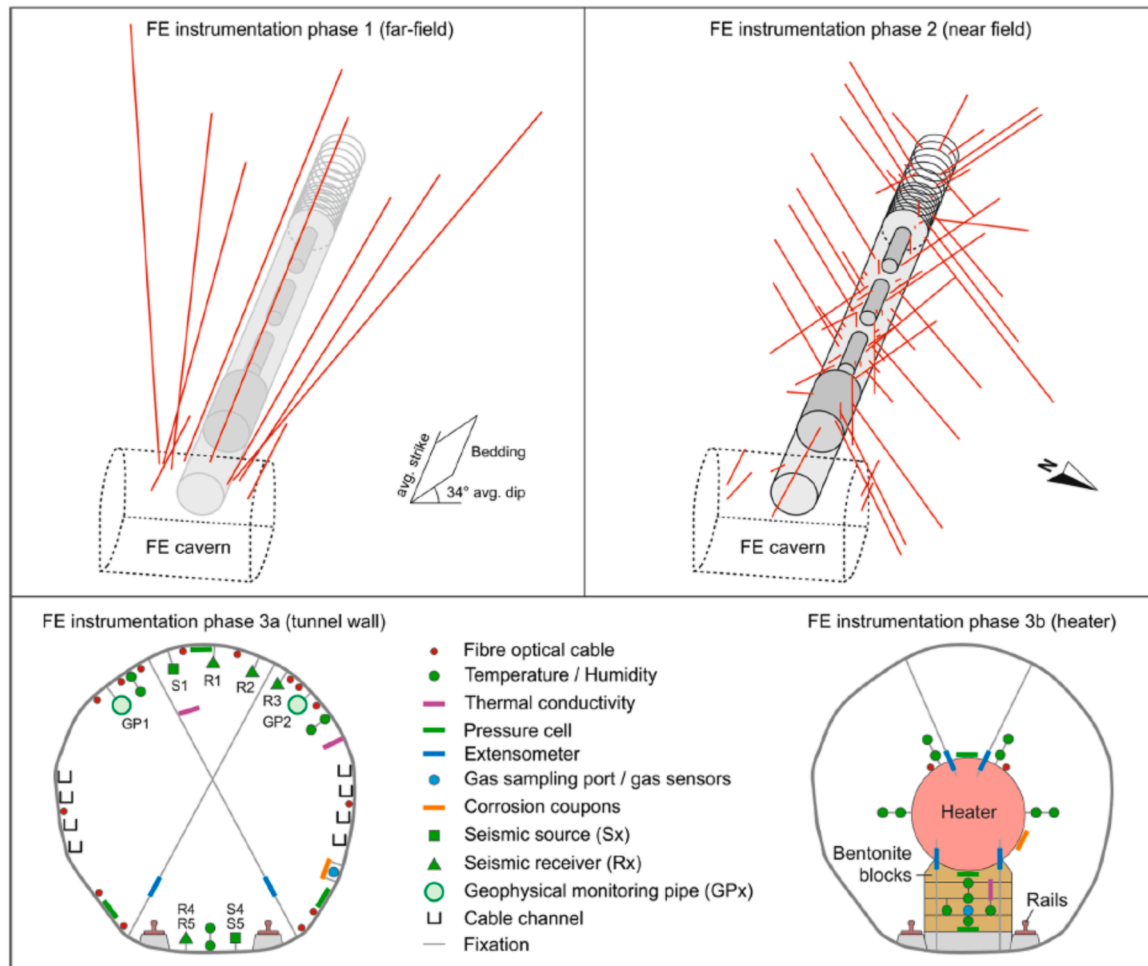


Fig. 2. Instrumentation of the FE-Experiment in and around the FE tunnel²⁴.

Table 2
Heating schedule²⁴.

	Date	Power (W)
Heater 1	15.12.2014	500
	13.01.2015	1000
	16.02.2015	1350
Heater 2	17.02.2015	1350
Heater 3	18.02.2015	1350

Model development

Both the conceptual model and the governing equations are well documented in e.g.¹³ and Seyedi et al.³⁸ and they are summarized here based on that literature.

Conceptual model

The heat emitted by the radioactive waste (or heaters in the case of the FE experiment) propagates through the backfilled tunnel and into the Opalinus Clay. The thermal conductivity of the bentonite blocks is roughly twice that of granular bentonite material (Fig. 3). Thus, it is

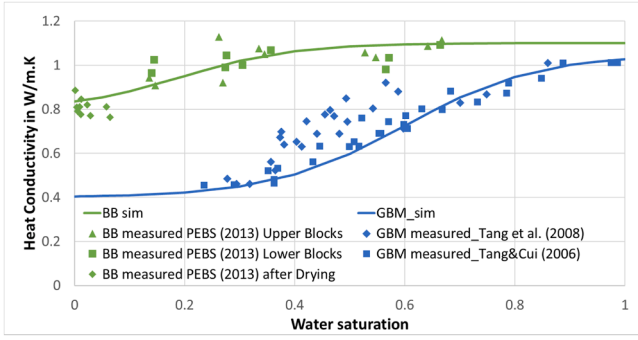


Fig. 3. Comparison of measured values of heat conductivity for Granular Bentonite Mixture (GBM) and Bentonite Blocks³⁰.

expected to observe a higher heat flux vertically downwards through the bentonite blocks. Within the Opalinus Clay, heat transport will be affected by the anisotropy of the rock, with heat conducted faster parallel to bedding than perpendicular, resulting in higher temperatures in the direction parallel to bedding than perpendicular at equal radial distance from the heaters.

The temperature increase in the Opalinus Clay heats both the pore water and the rock matrix, which remain in thermal equilibrium with each other. The heating causes thermal expansion of both the solid matrix of the rock and the pore water. Under incomplete confinement, the expansion of the solid matrix results in an increase in porosity, but this is smaller than the increase in water volume and therefore a pore pressure is generated.¹⁴ Porewater in the Opalinus Clay moves along the resulting pressure gradients but this process is slow because of the low permeability. The increased pressure causes a reduction in effective compressive stresses within the rock matrix, which could lead to increased likelihood of fracturing in the Opalinus Clay.

Close to the tunnel porewater drains into the tunnel supported by the suction of the bentonite. The Opalinus Clay close to the tunnel wall, the Engineered Damage Zone (EDZ), will have altered physical properties due to damage associated with stress redistribution as the tunnel is constructed. Previous modelling studies found that representing the EDZ as a zone of weaker rock with higher permeability improved the model representation of drainage to the tunnel.^{13,38,41}

Governing equations

The teams were asked to represent thermal, hydraulic (including vapour) and mechanical (linear elastic) processes in their models. A set of governing balance equations were defined (see Table 8 for symbol definitions) to represent conservation of heat (Eq. 1), mass of water (Eq. 2) and momentum (Eq. 3):

$$C_p \rho_b \frac{\partial T}{\partial t} - \nabla \cdot (\lambda \nabla T) = Q_H \quad (1)$$

$$\frac{1}{V} \frac{d(V \phi s \rho_w + V \phi (1-s) \rho_v)}{dt} = -\nabla \cdot (\rho_w \mathbf{q}_l + \rho_v \mathbf{q}_v) \quad (2)$$

$$\rho_b \frac{\partial^2 \mathbf{u}}{\partial t^2} = \nabla \cdot \bar{\sigma} + \rho_b \mathbf{g} \quad (3)$$

Teams were requested to implement the equations as closely as possible to the specification for the benchmark cases to eliminate differences between models. This was not always possible for all the teams, and key differences are captured in Table 3. The equations defined here are based on Biot's theory of consolidation³ and extended to thermo-hydroelasticity and partially saturated porous media through Philip and De Vries³¹ theory for heat and moisture flow in soil.^{28,36} In this theory, three phases (solid, liquid and gas) are present, but the gas pressure is constant and equal to atmospheric pressure in a static gas phase, thus

conservation of mass of air is not included. The heat balance equation (Eq. 1) assumes heat transport dominated by conduction meaning that heat advection with liquid water or vapour is neglected. This approach has been extensively applied in previous DECOVALEX phases for modelling THM processes in clay-based materials, with validation against laboratory and field experiments, including Kamaishi Mine heater test in Japan,³⁷ FEBEX at Grimsel in Switzerland,¹ and HE-E experiment at Mont Terri Laboratory.¹³ A detailed comparison with a full multiphase flow implementation was performed as part of this task³² and no significant differences were found under the assumption that the air retains a certain degree of mobility.

To enable accurate code-to-code verification and benchmarking teams were given a detailed specification for constitutive models and material properties. Of particular interest is the anisotropic structure of the Opalinus Clay, which results in anisotropic properties in the bedding parallel and bedding perpendicular direction. This affects thermal conductivity, permeability and Young's Modulus.

The thermal conductivity (Eq. 4), heat capacity (Eq. 5) and bulk density (Eq. 6) are assumed to depend on saturation:

$$\lambda = \lambda_{dry} + (\lambda_{sat} - \lambda_{dry}) \cdot s \quad (4)$$

$$C_p = C_p^{solid} \cdot (1 - \phi) \cdot \frac{\rho_g}{\rho_b} + C_p^{water} \cdot \phi \cdot s \cdot \frac{\rho_w}{\rho_b} \quad (5)$$

$$\rho_b = \rho_g (1 - \phi) + \rho_w \phi s + \rho_v \phi (1 - s) F_v + \rho_a \phi (1 - s) (1 - F_v) \quad (6)$$

Eq. 4 was used for the benchmark simulation whereas teams were free to adapt the equation for modelling the FE experiment (see Fig. 3). Liquid water flow is governed by Darcy's law as function of the liquid pressure gradient for both saturated and unsaturated conditions:

$$\mathbf{q}_l = -\frac{\kappa}{\mu} (\nabla p - \rho_w \mathbf{g}) \quad (7)$$

Vapour movement is assumed to results from diffusive processes due to density gradients in the static gas phase rather than moisture advection with bulk gas flow:

$$\mathbf{q}_v = -D_v \left(\nabla \left(\frac{\rho_v}{\rho_v^{sat}} \right) \right) \quad (8)$$

Capillary pressure and relative permeability are given by van Genuchten's equations:

$$s_c = \left(\frac{1}{1 + (P_e P_c)^n} \right)^m \quad (9)$$

$$\text{where } m = 1 - \frac{1}{n}, s_c = \frac{s - s_r}{s_{max} - s_r}$$

$$k_{rel} = s_c^{\frac{1}{2}} \left[1 - \left(1 - s_c^m \right) \right]^2 \quad (10)$$

Water density (ρ_w) is assumed to depend on both pore pressure and temperature according to

$$\rho_w = \rho_w^0 \frac{1}{\exp(\alpha_w (T - T_0)) \exp(\beta_w (P_0 - P))} \quad (11)$$

where α_w is the linear thermal expansion coefficient of water, specified as a constant for the purpose of benchmarking.

Temperature dependent water viscosity³⁹ is given by:

$$\mu = 4.2844 \cdot 10^{-5} + \left[0.157 \left(\frac{T - 273.15[K]}{K} + 64.993 \right)^2 - 91.296 \right]^{-1} [\text{Pa} \cdot \text{s}] \quad (12)$$

Vapour density is given as the product of saturated vapour density²² and relative humidity according to Kelvin's relation^{10,31}:

Table 3

Narrative descriptions of the differences between the models and the benchmark specification for each team, and how each team changed the equations when moving to modelling the FE experiment.

Team	Key differences for benchmark	Changes made for FE experiment
BGE/ TUBAF	Included vapour transport into heat balance Generalised energy balance equations ³² Additional porosity dependence on compressibility and thermal expansion of solid grains.	Vapour diffusion models dependent on temperature, saturation and tortuosity. Non-linear temperature and pressure dependent water density
BGR/ UFZ	Saturation dependent vapour diffusivity Explicitly considers gas phase, evaporation and advection of gas (including vapour) More general energy balance equations ³² Additional porosity dependence on compressibility and thermal expansion of solid grains.	
CAS	Porosity and saturation dependent thermal conductivity Linear water density model Water mass balance includes temperature dependent terms Additional porosity dependence on thermal expansion of solid grain.	
DOE/ LBNL	Solid grain thermal expansivity and compressibility considered.	Used steam tables for temperature and pressure dependent water properties. Used own choice of retention curve for bentonite and thermal conductivity for bentonite blocks. Include shotcrete with low permeability. Reduce rock permeability.
ENSI	Included vapour transport into heat balance Generalised energy balance equations ³²	Vapour diffusion models dependent on temperature, saturation and tortuosity. Non-linear temperature and pressure dependent water density
GRS KAERI/ KIGAM	Temperature dependent vapour diffusivity The volumetric strain is scaled by Biot coefficient in the porosity equation. Pore compressibility was used to calculate change in storage.	Used own choice of retention curves and permeabilities for Opalinus Clay and bentonite materials.
NWMO	Porosity equation includes pressure and temperature dependent effects as well as volumetric strain. ¹⁶	Used own choice of thermal conductivity as a function of saturation for granular bentonite and for bentonite blocks ¹⁷
DOE/ SNL	TH only model Constant heat capacity ¹⁸ equation of state for water viscosity and vapour density	Water and steam properties obtained from equation-of-state. ²⁰ Vapor diffusion coefficient calculated using the equation by Vargaftik ⁴² and Walker et al. ⁴³ Additional equations for thermal conductivity ⁴⁰ and pore compressibility ¹⁸ used.

$$\rho_v = RH \cdot \rho_v^{sat} \quad (13)$$

$$\rho_v^{sat} = 10^{-3} \text{kgm}^{-3} \cdot e^{\left(\frac{19,891 - 4975.9 \frac{[K]}{T}}{T} \right)} \quad (14)$$

$$RH = e^{p/(\rho_w RT/M_w)} \quad (15)$$

For the mechanical constitutive model, the total stress is related to strain through an elastic stiffness matrix, also taking into account the pore pressure. The convention of positive stress is tension is used here:

$$\bar{\sigma} = \bar{C} : (\epsilon - \epsilon_{th}) - B \max(p_w - p_w^0, 0) I \quad (16)$$

Thermal expansion is given by:

$$\epsilon_{th} = \alpha_T (T - T_0) \quad (17)$$

Equations to calculate porosity were specified for cases where only TH processes are simulated and also for cases where THM processes are simulated. As the TH cases are not discussed in detail in this paper, the porosity equations are only given for the THM cases. Assuming the solid grains are incompressible, the porosity can be calculated as:

$$\phi = 1 - \frac{(1 - \phi_0)}{(1 + \epsilon_{vol})} \quad (18)$$

Some teams used alternative formulations for porosity, including the expansion of the solid grains as discussed in Table 3.

When moving from the benchmarking exercises to the representation of the FE experiment, teams were encouraged to move away from the specification and make their own choices for material models, parameter values and the components of the system that are represented. The key changes to equations for the FE model are captured in Table 3.

Benchmarking models

The aim of benchmarking is to build confidence in each team's model

and its ability to represent processes that are important in modelling the FE experiment. With a tightly defined specification, the aim of this exercise is for all teams to achieve the same results, or to understand the cause of the difference. The aim of this step is not to produce the best model possible of the FE experiment, so the specification contains known simplifications to aid model comparison rather than try to replicate results from the FE experiment.

A series of 2D benchmark cases were designed, starting with only thermal processes, then adding hydraulic processes and finally full THM coupling. Additional simplifications were introduced to determine the cause of differences between models, including removing all bentonite from models and replacing it with Opalinus Clay, and setting the permeability of the Opalinus Clay to zero. In this paper, we only report the results from the final 2D THM benchmark case.

The benchmark cases are based on a 2D slice through the centre of the middle heater in the FE experiment (Fig. 4), including representation of the Opalinus Clay and two types of bentonite. The teams were asked to represent the heater as a hole and a set of boundary conditions. The model domain size was consistent between the teams at 50 m x 50 m. The locations of the points for data output are shown in Fig. 4 and were used for comparison between the teams.

Alongside the material models and parameters provided to the teams, the initial and boundary conditions were given (Table 4).

The results for the T and the TH case (not presented) show a very close agreement between all teams for temperature and pressure. For the THM case there are some differences for the pressure results. All teams show a good agreement for the temperature (Fig. 5). For the pore pressure (Fig. 6) we notice some differences between the teams especially at the end of the simulation time but still there is a close agreement between all but one team. The results of KAERI in Fig. 6 show a different pressure evolution over time because the model used a pore compressibility model linking porosity to pore pressure.

The close agreement between the teams provided confidence that the relevant THM processes are correctly implemented in the various simulators. This was supported by comparison to analytical solutions of

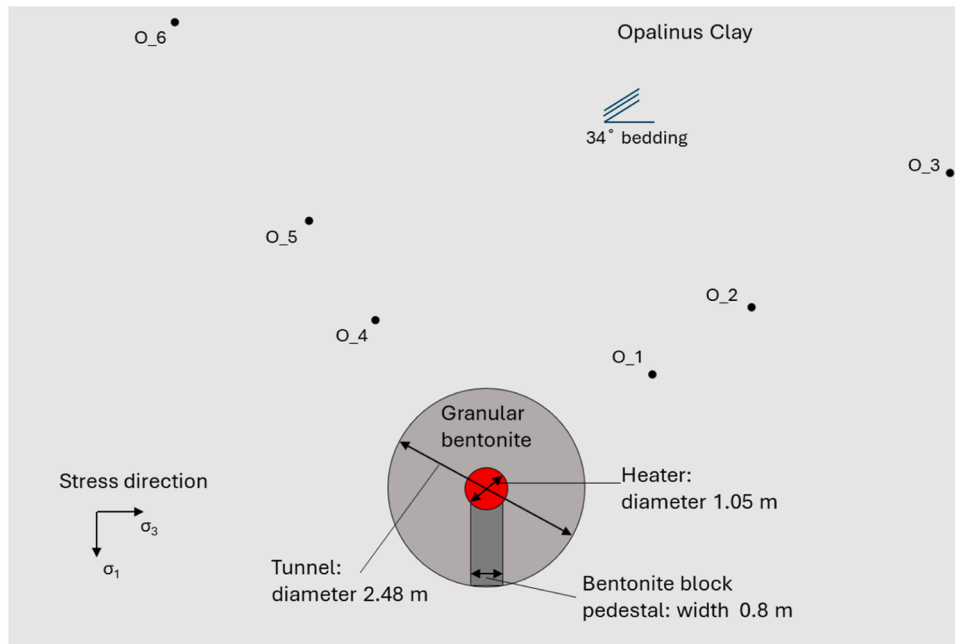


Fig. 4. Model domain around the FE tunnel showing the tunnel with heater and bentonite backfill. The locations at which model outputs are required are also shown (O_1 to O_6). The size of model domain recommended for benchmarking is 50 m x 50 m.

Table 4
Initial and boundary conditions for the benchmark models.

	Initial	Heater boundary	Outer boundaries
Thermal	15C	88.969 W/m2	No heat flow
Hydraulic	OPA: 2 MPa pore pressure, saturation of 1 Granular bentonite: water content 5 wt% Bentonite blocks: water content 18 wt%	No flow	No flow
Mechanical	$\sigma_1 = \sigma_{zz} = 6.5$ MPa $\sigma_3 = \sigma_{xx} = 2.5$ MPa	Zero displacement	Zero displacement

simplified test cases (not shown). A number of teams carried out additional sensitivity tests and showed that the domain size was affecting the results and a larger domain size was required for the mechanical calculations. They also showed that the assumption of a constant thermal expansion of water was inaccurate and that including a temperature-dependent thermal expansion coefficient of water was an important consideration for representing the experiment.

Modelling the FE experiment

Moving away from benchmarking and on to modelling the FE experiment, the teams were provided with a subset of data from the experiment and asked to produce a model that could reproduce the data. The teams had more freedom to change the modelling approach including whether they started the simulation with the excavation, the ventilation or the heating phase, the technical features they would like to consider including shotcrete or excavation damaged zone (EDZ) and which material models and parameters they allocated to the different materials. Teams also choose different domain sizes and boundary conditions for the heater.

The teams were provided with the experiment schedule (Table 5). Time zero was defined as the onset of heating on 15th December 2014.

Two teams (ENSI and NWMO) included the excavation of the tunnel

in their model, with ENSI simulating excavation over 100 days and NWMO assuming instantaneous excavation. Both models assumed linear elasticity in the Opalinus Clay, so the excavation did not produce a damage zone but changed the stresses and pore pressures in the Opalinus Clay. BGR/UFZ also investigated simulation of excavation using a linear elastic model but concluded that it didn't explain satisfactorily the early liquid pressure drop from the initial 2 MPa visible in the field measured right after excavation. Therefore, they chose to postulate an initial pressure field (with lower pressures near the tunnel and with an elliptical shape to take into account the anisotropy in the permeability) to match measurements at $t = -900$ days ($t = 0$...start of heating of the first heater), while acknowledging that processes taking place before that could not be explained by the chosen modelling approach. During the ventilation phase, no large pressure changes were measured in the host rock.

All teams represented the ventilation phase of the FE experiment. Six teams (BGR/UFZ, BGE/TUBAF, CAS, ENSI, GRS and DOE/LBNL) chose to represent the ventilation of the tunnel by applying a suction pressure as a boundary condition on the tunnel wall. The suction pressure was related to the relative humidity in the tunnel by the Kelvin equation. Teams chose different suctions as shown in Table 6 and ventilated the tunnel for different amounts of time. Two teams (KAERI/KIGAM and NWMO) applied a boundary condition of atmospheric pressure to the tunnel wall and one team (DOE/SNL) applied an atmospheric boundary condition at the end of the tunnel and assumed the tunnel was filled with high permeability high porosity material. Most teams that applied a suction boundary on the tunnel surface used relatively low values of suction that do not correspond to the measured relative humidity in the tunnel, but it was needed to reproduce the slow and light decrease in porewater pressure in the Opalinus Clay. This could be explained by the presence of a boundary layer at the tunnel wall where the relative humidity was higher and suction was thus lower (consistent with the finding of⁶).

Three teams included an EDZ (BGE/TUBAF, KAERI/KIGAM and NWMO). BGE/TUBAF assumed different thicknesses in different directions related to the bedding in the Opalinus Clay (elliptical cross section) and depending on the presence or absence of shotcrete at the tunnel wall. The other two teams used uniform thicknesses (Table 6). The EDZ was modelled with higher permeability and lower Young's

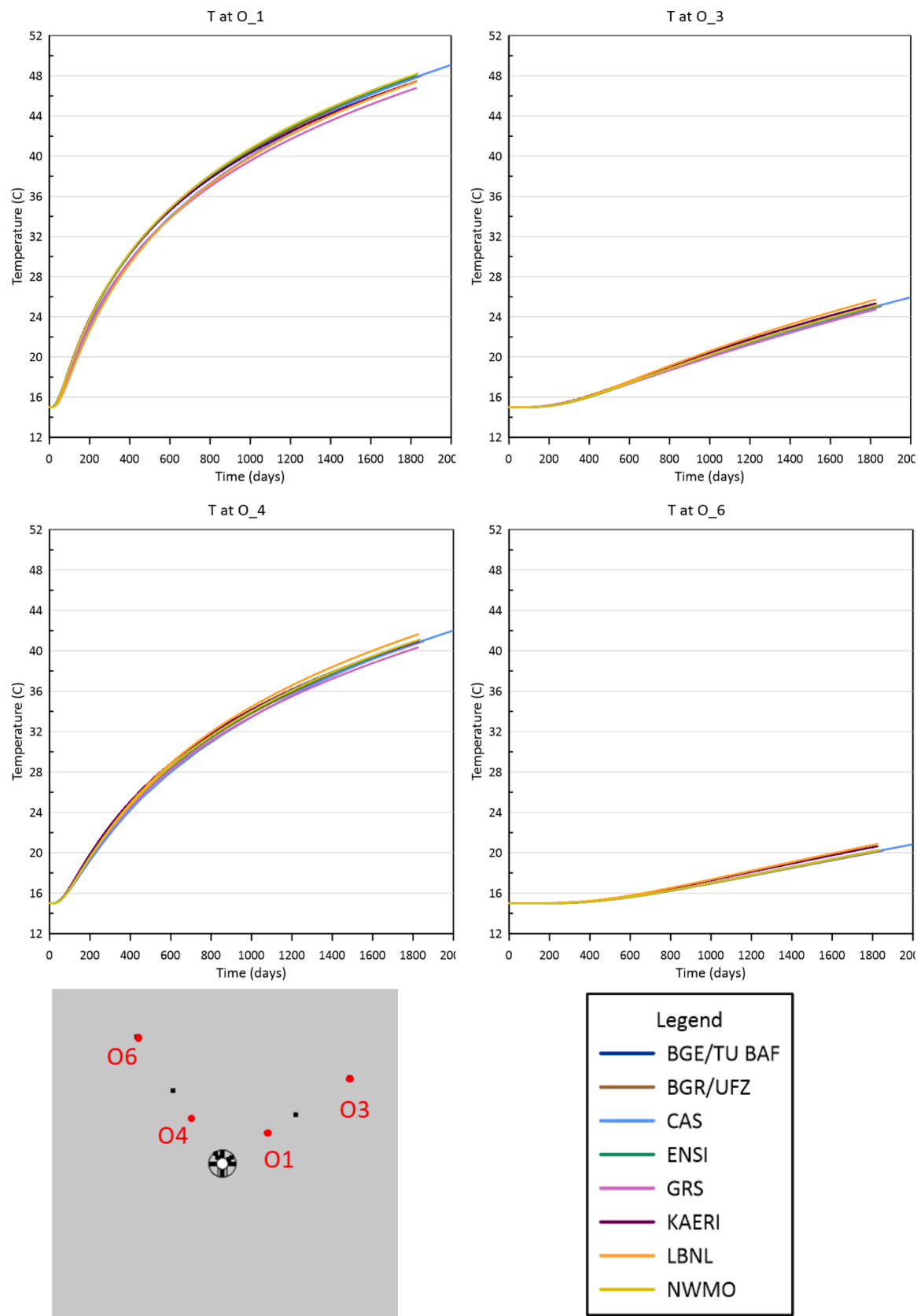


Fig. 5. Selected temperature results for locations in the Opalinus Clay from Step 0c.

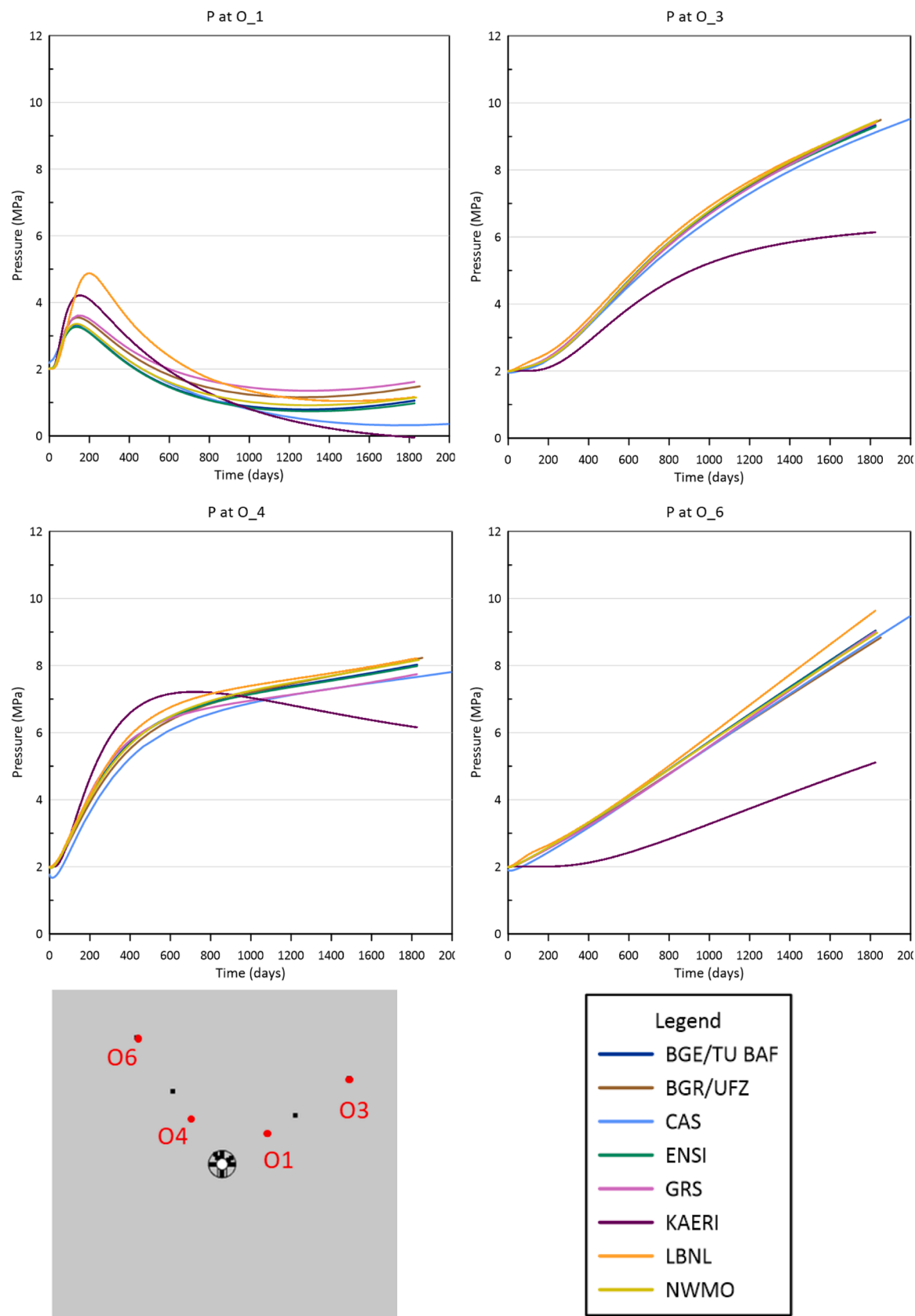


Fig. 6. Selected pressure results for locations in the Opalinus Clay from Step 0c.

Table 5
Experiment schedule²⁷.

	Date
Start of excavation	26.04.2012
End of excavation	11.07.2012
Start of backfilling	04.07.2014
Backfill around H1	23.10.2014
Backfill around H2	20.11.2014
Backfill around H3	21.01.2015
Concrete plug Emplacement	17.03.2015
Start of heating	15.12.2014

Modulus than the Opalinus Clay (Table 6). Most teams chose to represent shotcrete lining the tunnel with thicknesses around 20 cm, but had different values for shotcrete permeability, with a range of 10^{-17} to 10^{-21} m².

Teams updated the material models (Table 6) and parameters that they were using (Table 7). For some teams this was based on literature values and previous work. Two teams (NWMO and BGE/TUBAF) undertook extensive parameter calibration testing a wide range of parameter combinations and making a judgement (by eye for NWMO and using a numerical scoring system for BGE/TUBAF) to choose the best parameter set.

The heater representations, the domain size and the number of nodes/elements of the grids chosen by the teams are presented in Table 6.

Results for the FE experiment

The temperature evolution in the Opalinus Clay (Fig. 7), shows a good match with the measurements and a small spread between the teams at all sensors. Some teams underpredicted the temperature evolution parallel to bedding slightly. The initial temperature that CAS have used was higher than the agreed value.

For the temperatures within the bentonite, the majority of teams show good agreement with the data, particularly at the heater surface (Fig. 8). BGR/UFZ and CAS have consistently too high temperatures in the bentonite (by 10–30°C) while GRS has too low temperature in the bentonite blocks (by 5–10°C).

The water pressure evolution in the Opalinus Clay is shown in Fig. 9. Most team results agreed nicely with the measurements whereas some

teams over- or underestimated the pressure increase due to heating for sensors close to the heater and several teams overestimate the drainage effect slightly compared to measurements. CAS was not able to reproduce the evolution of porewater pressures and reported very high water pressure values (up to 4.5 MPa). The teams that reported pressure values during ventilation consistently found that the pressure decreased faster in the models than in the data.

The relative humidity in the bentonite is consistent between the teams in terms of the shape of curve although the final relative humidity varies by 20 %. All teams captured the initial increase in relative humidity at the onset of heating and the subsequent decrease due to dry out (Fig. 10).

Prognosis of the future evolution following changes of the heat power in the FE experiment

The course of the FE experiment was changed in March 2023 with an increase of the heat power and a year later with the decrease of the heat power, see Fig. 11. The calibrated models were used to predict how temperature and pressure in the Opalinus Clay will change through time beyond the available data following those disturbances of the experiment regime.

The modelled temperatures (Fig. 12) are all within 2°C of each other over the 6000-day time period modelled, and there is no evidence of the models diverging thereafter. The differences between the models arise within the first 2000 days and are then stable. This suggests that the differences are caused by the thermal conductivities. The numerical results at the two sensors closest to the tunnel (BFEA002_TEM_03 and BFEA005_TEM_03) show a short-term peak in temperature between 3000 and 4000 days related to the increase in heater power during 2023 and early 2024. Further into the Opalinus Clay (BFEA003_TEM_03 and BFEA005_TEM_03), both the increase in heater power and its subsequent decrease in March 2024 have a less prominent impact on the predicted temperature evolution. The modelled pressures in the Opalinus Clay (Fig. 13) are within 2 MPa of each other over the 6000-day time period, for the sensors closest to the heater. However, models do have different rates of pressure evolution at the end of the model run and some teams overpredicted and some teams underpredicted the pressure evolution during the first 2000 days for which data were available. For the sensors further away from the heater, the pressure difference is less than 1 MPa, providing a better estimate of the pressure evolution over the next few

Table 6
Heater representation, domain sizes, boundary conditions used to represent ventilation of the tunnel, and additional materials chosen by teams.

Team	Heater representation	Domain size (m)	Number of nodes/elements	Water boundary condition	Modelled duration of ventilation	EDZ	Shotcrete
BGR/UFZ	Surface boundary	150 × 100 x 150 m ³	120,000/720,000	69.0 MPa suction at tunnel wall (60 % RH)	2.5 years	No	20 cm thick, permeability 1E–17 m ²
BGE/TUBAF	Solid volume	200 × 200 x 200 m ³	475313 / 2909597	57.2 MPa suction (65 % RH)	2.5 years	Parallel 2 m (0.8 m where shotcrete) Perp 2.5 m (1.0 m where shotcrete)	16–24 cm thick (changes along tunnel), Permeability 1.15E–17 m ²
CAS	Thin-walled cylinder	100 × 100 x 55 m ³		57.4 MPa suction (65 % RH)	2.5 yrs	No	No
DOE/LBNL	Solid volume	150 × 150 x 150 m ³	64,158 / 67,627	2 MPa suction	3 yrs	No	18 cm Thick, Permeability 3e–22 m ²
ENSI	Solid volume	150 × 100 x 150 m ³	99,397 / 597,634	2 MPa suction (98.5 % RH)	2.5 yrs	No	25 cm thick, Permeability 1.75E–19 m ²
GRS	Solid volume	100 × 100 x 49.3 m ³	29,895/27,910	2 MPa suction	2 years	No	No
KAERI/KIGAM	Surface boundary	100 × 60 x 100 m ³	44,586 / 85,080	Atmospheric pressure	3 yrs	1.8 m circle	20 cm, Permeability 1E–17 m ²
NWMO	Solid volume	100 × 90 x 100 m ³	140,888/269,525	Atmospheric pressure	3 yrs	3 m thick, Permeability 20x greater than Opa	No
DOE / SNL	Solid volume	50 × 50 x 50 m ³		Atmospheric pressure on tunnel end. Tunnel has permeability of 1E–12 m ² and porosity of 0.9	1000 days	No	30 cm thick, Permeability 2E–21 m ²

Table 7

Provided parameters based on^{5,7,11} and²⁶ as well as the range of parameters used by the teams for Step 2.

Input parameters		Symbol	Unit	OPA	Teams parameter range	GBM	Teams parameter range	Bentonite blocks	Teams parameter range	Shotcrete/Concrete	Teams parameter range
Thermal parameters	Dry thermal conductivity parallel and perpendicular to bedding	$\lambda_{\text{dry},\parallel}$	W/mK	2.4	1.71–2.5	0.35	0.23–0.4	0.26	0.26–1	0.1 ^a	0.1–1.7
		$\lambda_{\text{dry},\perp}$	W/mK	1.3	0.94–1.3						
	Saturated thermal conductivity parallel and perpendicular to bedding	$\lambda_{\text{sat},\parallel}$	W/mK	2.4	1.71–2.5	1.2	1.0–1.16	0.96	0.87–1.3	0.3 ^a	0.3–1.7
		$\lambda_{\text{sat},\perp}$	W/mK	1.3	0.94–1.3						
Hydraulic parameters	Solid specific heat capacity	c_s	J/kgK	995	995	800	800	800	750–800	750	750
	Dry Bulk Density	ρ_{bulk}	kg/m ³	2340	2340	1490	1490	1690	1690	1725	1725–2300
	Porosity	ϕ	-	0.13	0.13–0.17	0.331	0.331–0.45	0.331	0.331–0.36	0.25	0.23–0.25
	Intrinsic permeability	$k_{i,\parallel}$	m ²	5.0E–20	6.2E–21–4E–20	3.5E–20	3.5E–21–3.5E–20	1.0E–22	1E–22–2E–21	1.00E–19	2E–21–1.15E–17
		$k_{i,\perp}$		1.0E–20	2.5E–21–3E–20						
	van Genuchten Entry Pressure	p_e	MPa	20.0	12–20	28.6	10.0–30.0	30	21.9–64	1	1.0–20.0
	van Genuchten n	n	-	2.5	1.4–2.5	2.0	1.67–3.0	1.67	1.43–1.67	1.49	1.49–2.5
	van Genuchten maximum water saturation	s_{max}	-	1.0	1.0	1.0	1.0	1.0	1.0	1	1
	van Genuchten residual water saturation	s_r	-	0.0	0.0	0.0	0.0	0.0	0.0	0.01	0.01
	Pore compressibility	c_{pore}	1/Pa	-	"-" or 5.13E–10–8.66E–10	-	"-" or 1.05E–7–1.05E–6	-	"-" or 1.13E–7–2E–6	-	"-" or 3.5E–12–1.4E–10
Mechanical parameters	Young’s modulus	$E_{\parallel}$	MPa	8000	1600–8000	18	18	24	24	20000	20000–21500
		$E_{\perp}$		4000	800–4000						
	Shear modulus	$G_{\perp}$	MPa	3500	880–3500	-	0	-	0	-	0
	Poisson ratio	$\nu_{\parallel}$	-	0.35	0.35	0.35	0.35	0.2	0.2	0.15	0.15
$\nu_{\perp}$			0.25	0.25							
Fluid parameters	Linear thermal expansion	α_T	1/K	1.5E–05	1E–5–1.5E–5	3.0E–06	3.0E–06	3.0E–06	3.0E–06	1.50E–05	1.50E–05
	Biot coefficient	α	-	1	0.6–1	1	1	1	1	1	1
	Reference water density	$\rho_{\text{fluid,ref}}$	kg/m ³	1000							
	Fluid compressibility	c_{fluid}	1/Pa	4.65E–10							
	Linear thermal expansion water	α_w	1/K	4.00E–04							
	Vapour diffusivity (vapour in air)	D_v	m ² /s	2.42E–05							

^a These values from the specification were later changed to 1 and 1.7 W/mK

Table 8

Definition of symbols used in this paper.

Symbol	Name	Units	Definition
C_p	Specific heat capacity	J/kg/ K	Eq. 5
ρ_b	Bulk density	kg/m ³	Eq. 6
T	Temperature	K	Dependent variable
λ	Thermal conductivity	W/m/ K	Eq. 4
Q_H	Heat source	W /m ²	Boundary condition
ϕ	Porosity	-	Eq. 18
s	Saturation	-	Calculated
ρ_w	Density of water	kg/m ³	Eq. 11
ρ_v	Density of vapour	kg/m ³	Eq. 13
V	Volume	m ³	Calculated
q_l	Liquid flux	kg/ m ² /s	Eq. 7
q_v	Vapour flux	kg/ m ² /s	Eq. 8
κ	Permeability tensor	m ²	Table 0–6
μ	Liquid viscosity	Pa.s	Eq. 12
p	Liquid pressure	Pa	Dependent variable
$\bar{\sigma}$	Total stress tensor	Pa	Calculated
$\mathbf{u}$	Displacement vector	m	Dependent variable
$\mathbf{g}$	Gravity	m/s ²	9.81 m/s ²
λ_{dry}	Dry thermal conductivity	W/m/ K	Table 0–6
λ_{sat}	Saturated thermal conductivity ()	W/m/ K	Table 0–6
C_p^{solid}	Heat capacity of solid grains	J/kg/ K	Table 0–6
C_p^{water}	Heat capacity of water	J/kg/ K	Table 0–6
ρ_g	Density of grains	kg/m ³	Calculated from ρ_{bulk} Table 0–6 and porosity
D_v	Diffusivity of vapour	m ² /s	Table 0–6
ρ_v^{sat}	Saturated vapour density	kg/m ³	Eq. 14
P_c	Capillary pressure	Pa	Eq. 9
s_e	Effective saturation	-	Eq. 9
m	Van Genuchten parameter	-	Eq. 9
k_{rel}	Relative permeability (-)	-	Eq. 10
s_{max}	Maximum saturation (-)	-	Table 0–6
s_r	Residual saturation (-)	-	Table 0–6
P_e	Van Genuchten Entry pressure	Pa	Table 0–6
n	Van Genuchten parameter	-	Table 0–6
ρ_w^0	Reference water density	kg/m ³	Table 0–6
α_w	Linear thermal expansion coefficient of water	1/K	Table 0–6
C_{fluid}	Compressibility of water	1/Pa	Table 0–6
T_0	Reference temperature	K	293 K
p_0	Reference pressure	Pa	0.1 MPa
M_w	Molar mass water	g/mol	18 g/mol
R	Ideal gas constant	J/ K-mol	8.314 J/K-mol
RH	Relative humidity	%	Eq. 15
ϕ_0	Reference porosity	-	Initial porosity Table 0–6
ϵ_{vol}	Volumetric strain	-	Calculated
ϵ_{th}	Thermal strain	-	Eq. 17
B	Biot coefficient	-	Table 0–6
$\mathbf{I}$	Identity matrix	-	
$\mathbf{C}$	Stiffness matrix	1/Pa	Calculated
α_T	Coefficient of thermal expansion	1/K	Table 0–6

years.

The predictions for the relative humidity (Fig. 14) show a good agreement over time. The discrepancy between the team's predictions is

up to 20 %. Only one team predicted an increase of the relative humidity during the next 10 years.

Discussion

One of the motivations for the FE experiment was to provide data for validation of existing numerical models of coupled thermo-hydro-mechanical (THM) processes within the Opalinus Clay host rock, at the full scale for geological disposal.²⁴ The work presented in this paper has used the data from the FE experiment for this purpose. There is close agreement between the teams' model results and the measurements from the experiment for temperature and relative humidity. Whilst there is a more significant spread of model results for pore pressure within the Opalinus Clay, and associated less good agreement between the models and the measurements, the majority of the models do capture the broad trends of the observations, demonstrating that the first order processes are well captured in the models.

The comparison of porewater pressure evolution in Opalinus Clay shows larger discrepancies between measurements and models close to the tunnel contour. The models either overpredicted the pressure increase or overpredicted the drainage effect. With greater distance, the discrepancies diminish and at the sensors farthest away from heater the models tend to underpredict the pressure evolution. Teams that calibrated their model to the data were only able to achieve a very close fit between the model and the observations by using parameter values for the Opalinus Clay (particularly Young's Modulus) that are outside of the expected range for intact Opalinus Clay.

Following this observation, several teams experimented with using lower values of Young's Modulus in their model. Reducing the Young's Modulus values to 60 %, 40 % and 20 % of the original values given in Table 7 improved the fit of model pore pressure to experimental data at sensor locations closer to the tunnel (O2 and O5) but had less clear benefits at sensors further away from the tunnel (O4 and O7). Fig. 15 shows the effect on the porewater pressure with reduced Young's modulus. The teams reduced the values to 20 % resp. 25 % of the values given in Table 7 (1600 MPa parallel and 800 MPa perpendicular instead of 8000 MPa parallel and 4000 MPa perpendicular). While the selection of a value as low as 60 % of the original provided Young's modulus would be supported by findings of,⁴⁴ selecting an even lower value cannot be justified by tests in the laboratory but might be indicative of inelastic effects that cannot be covered in laboratory tests. These results suggest that the rock mass around the FE tunnel may behave in a less stiff manner closer to the tunnel and appear stiffer further away from the tunnel, probably caused by rock damage processes during and after the excavation. This is a consistent finding with previous authors,^{13,41} who concluded that explicit representation of the EDZ in models (e.g. as a weaker and higher permeability material than the intact rock) would affect the model representation of the build-up and dissipation of pressure within the host rock.

Three teams (BGE/TUBAF, KAERI and NWMO) tested this theory by including a simplified elastic representation of the EDZ in their models. There are a number of significant differences between these three models (e.g. BGE/TUBAF has a higher Young's Modulus for intact Opalinus Clay, KAERI's model uses pore compressibility), so it is not easy to draw firm conclusions, but there was no clear evidence that addition of an EDZ to the models improved pressure predictions (Fig. 9). The linear elastic models of the EDZ used by the three teams are a significant simplification on the material properties of the EDZ, which has an internal structure dependent on the anisotropy of both stress field and material properties and shows non-linearity with differing loading and unloading stiffness. Further investigation of representing these properties in a model could be helpful in understanding the pressure response of the Opalinus Clay.

It is also possible that there are other missing features and/or processes that would help the models represent the data better.⁸ Potential candidates could be better representation of the complexities within the

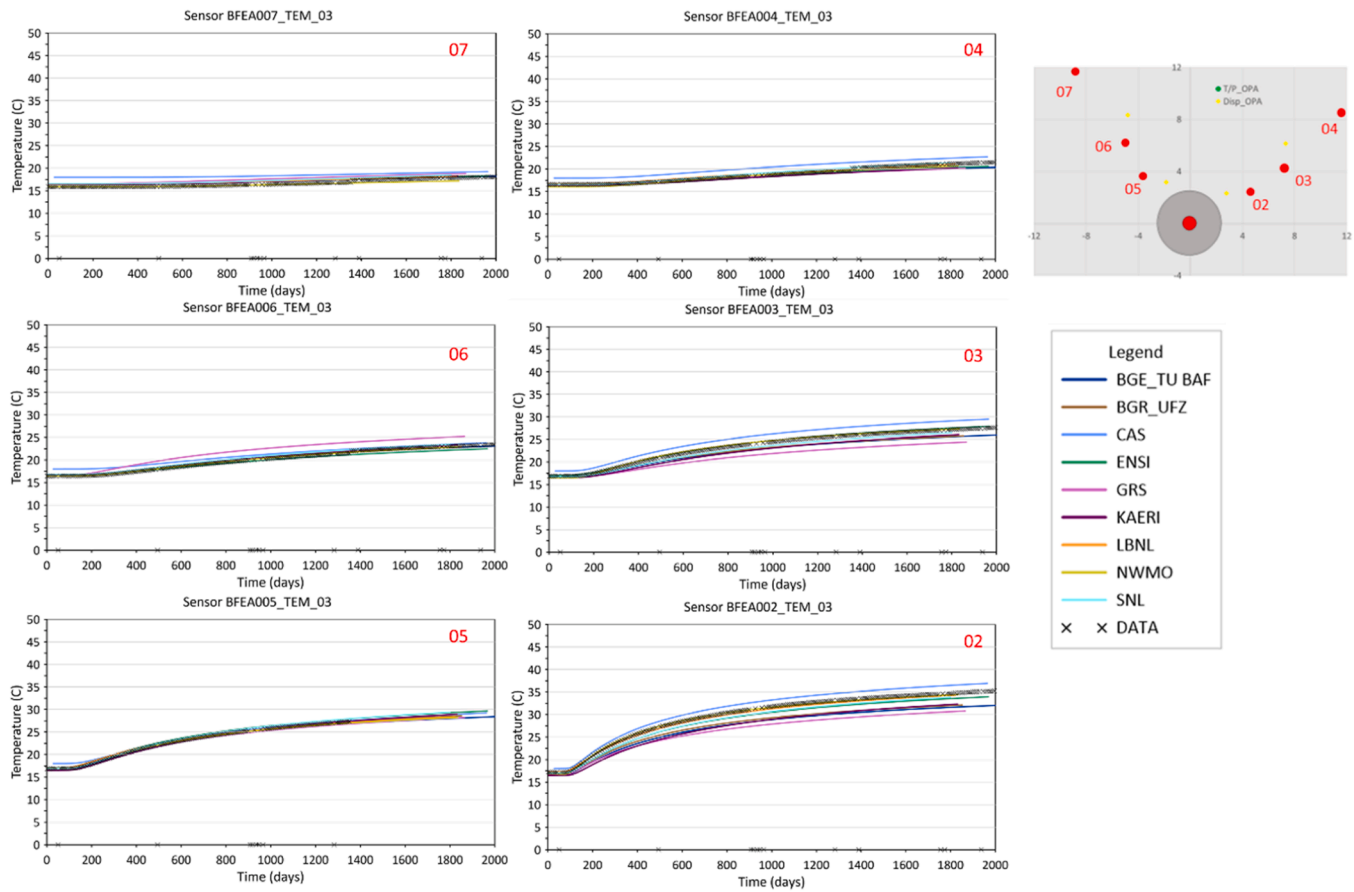


Fig. 7. Selected temperature results for locations in the Opalinus Clay from Step 2.

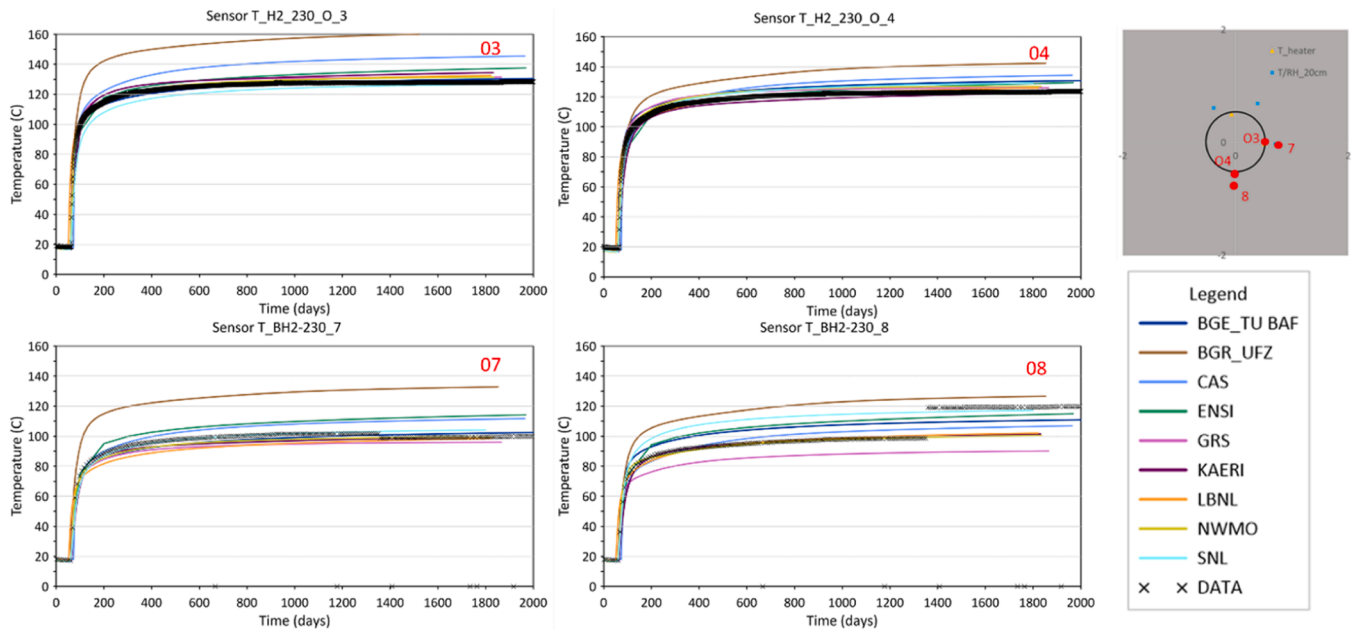


Fig. 8. Selected temperature results for locations in the bentonite from Step 2.

bentonite backfill, interactions with other experiments conducted in the Mont Terri URL, or variations in the reference stress or water pressure fields.

All teams chose to add a representation of the ventilation of the tunnel to their models for the FE experiment, with boundary conditions

to represent the ventilation ranging from suction applied to the tunnel wall (with values ranging from 2 MPa to 69 MPa) to atmospheric pressure at the boundary. Although teams applied a range of boundary conditions to the tunnel wall during ventilation, there was general agreement between the teams in that the data showed less of an effect of

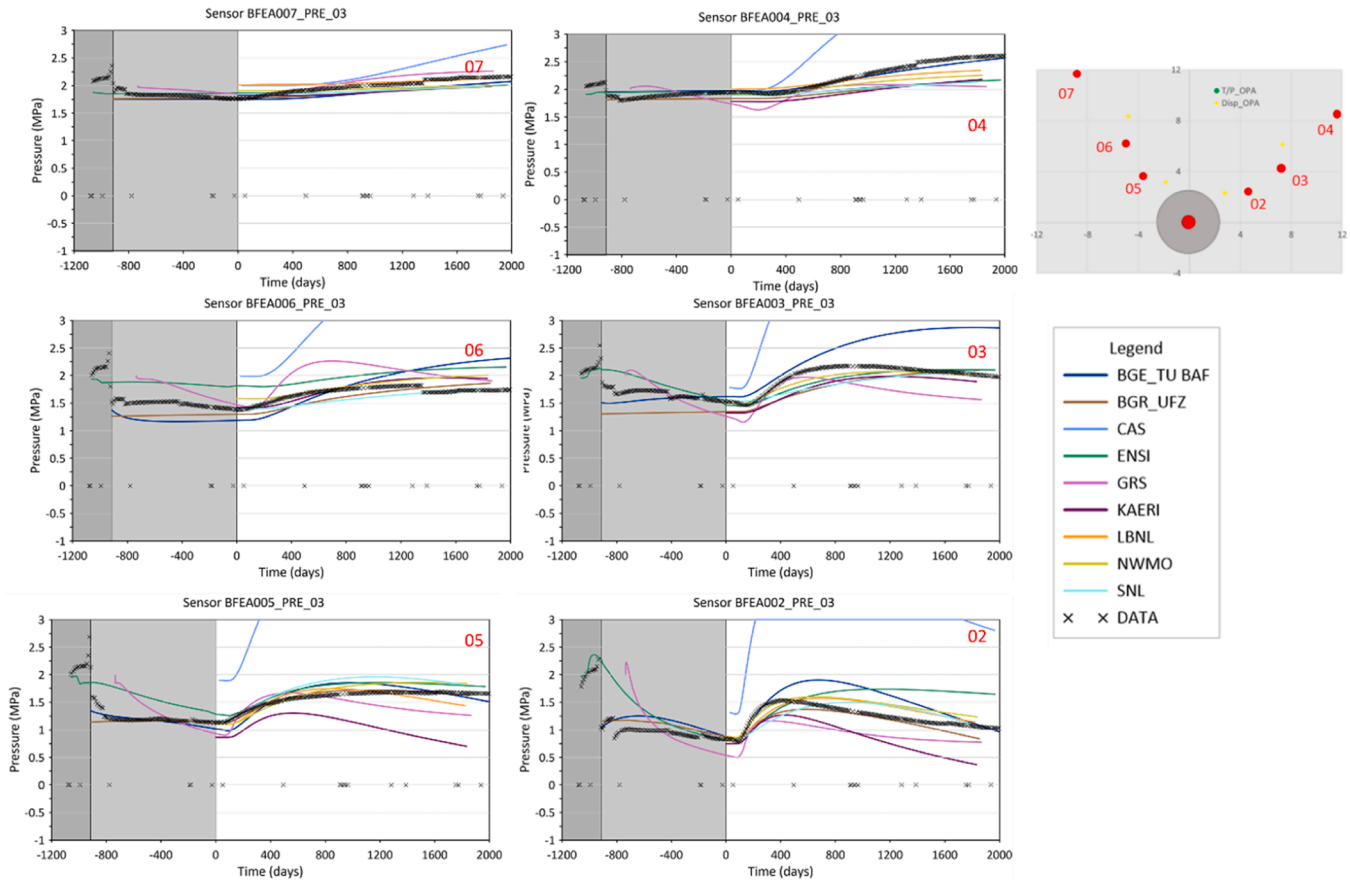


Fig. 9. Selected pressure results for locations in the Opalinus Clay from Step 2.

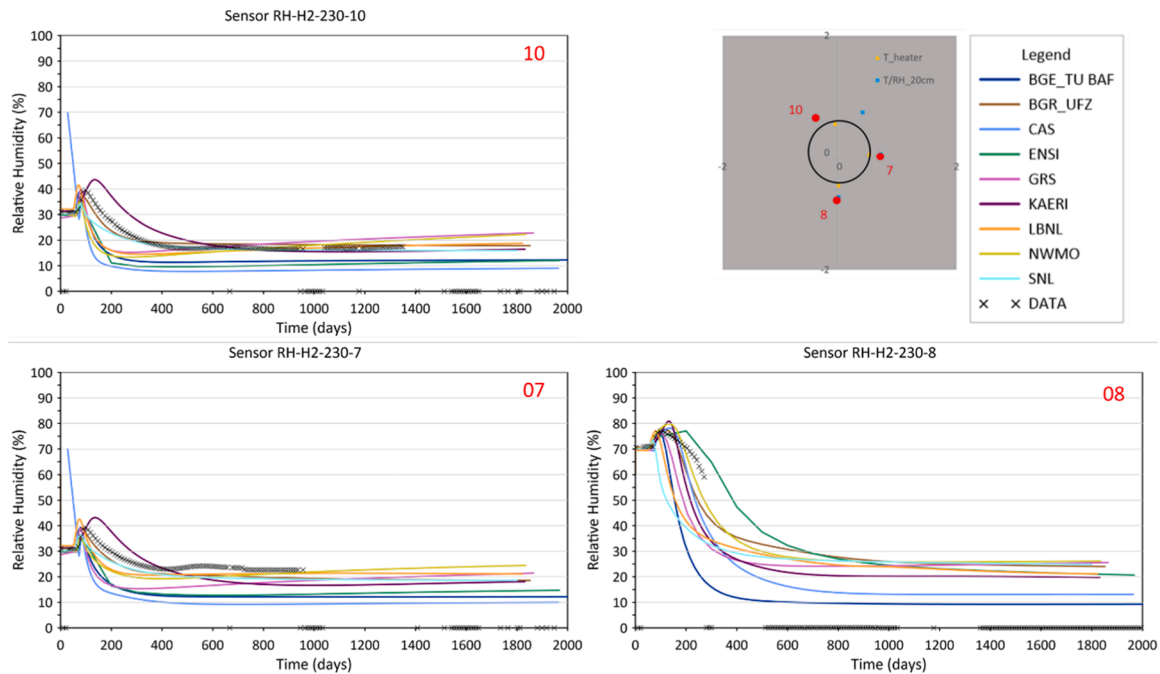


Fig. 10. Selected relative humidity results for locations in the bentonite from Step 2.

ventilation in the Opalinus Clay than the models. This is potentially because the Opalinus Clay was disturbed already during excavation and the teams have not modelled this in detail, or the variation in relative humidity across the tunnel (especially boundary layers around the

tunnel surface) may restrict water vapour loss from the tunnel walls.

An important use of THM models is in predicting future behaviour in both experiments and across whole repository systems. The models developed here were used to predict the temperature, pressure and

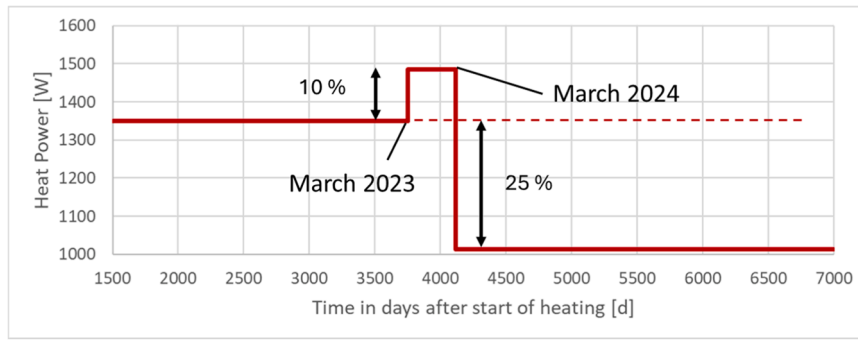


Fig. 11. Modified schedule of heating for the FE experiment in days after start of heating of the first heater.

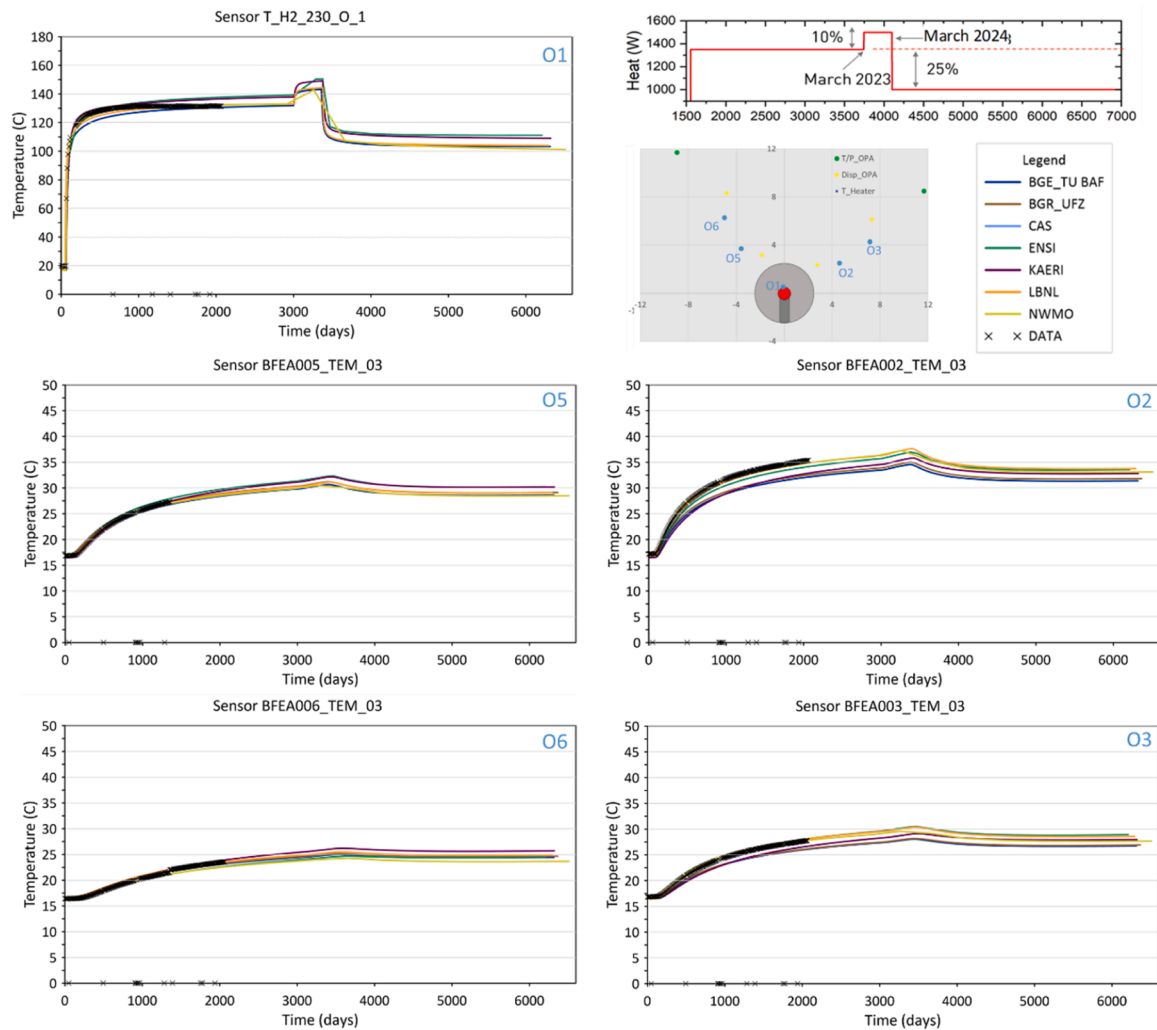


Fig. 12. Selected temperature results for locations directly at the heater and in the Opalinus Clay from Step 3.

relative humidity evolution of the FE experiment after the heating schedule was revised. The models all give very similar prediction of temperatures, indicating a small range of uncertainty for the expected future evolution. There is a larger range in predicted pore water pressures with the largest differences between the models found nearest to the tunnel, where the models were least consistent with the observations. In addition, the models are not all converging at the end of the model run, so the uncertainty within the predictions will increase with time. In the long term, the differences of the simulated maximum pressure between the teams are relatively small compared to the

increasing impact of the drainage effect of Opalinus Clay on the pore pressure with time. The predictions for relative humidity agree well over time, with all but one team showing resaturation of the bentonite is not predicted by the models over the next 10 years. It should be noted however that the detailed representation of resaturation of the bentonite was not the focus of this modelling work and simplified models were used. To get accurate predictions of bentonite resaturation, it would be beneficial to use models that represent swelling of the bentonite. This would allow to consider the change in pore size distribution of Bentonite due to swelling which affects the permeability of the bentonite.

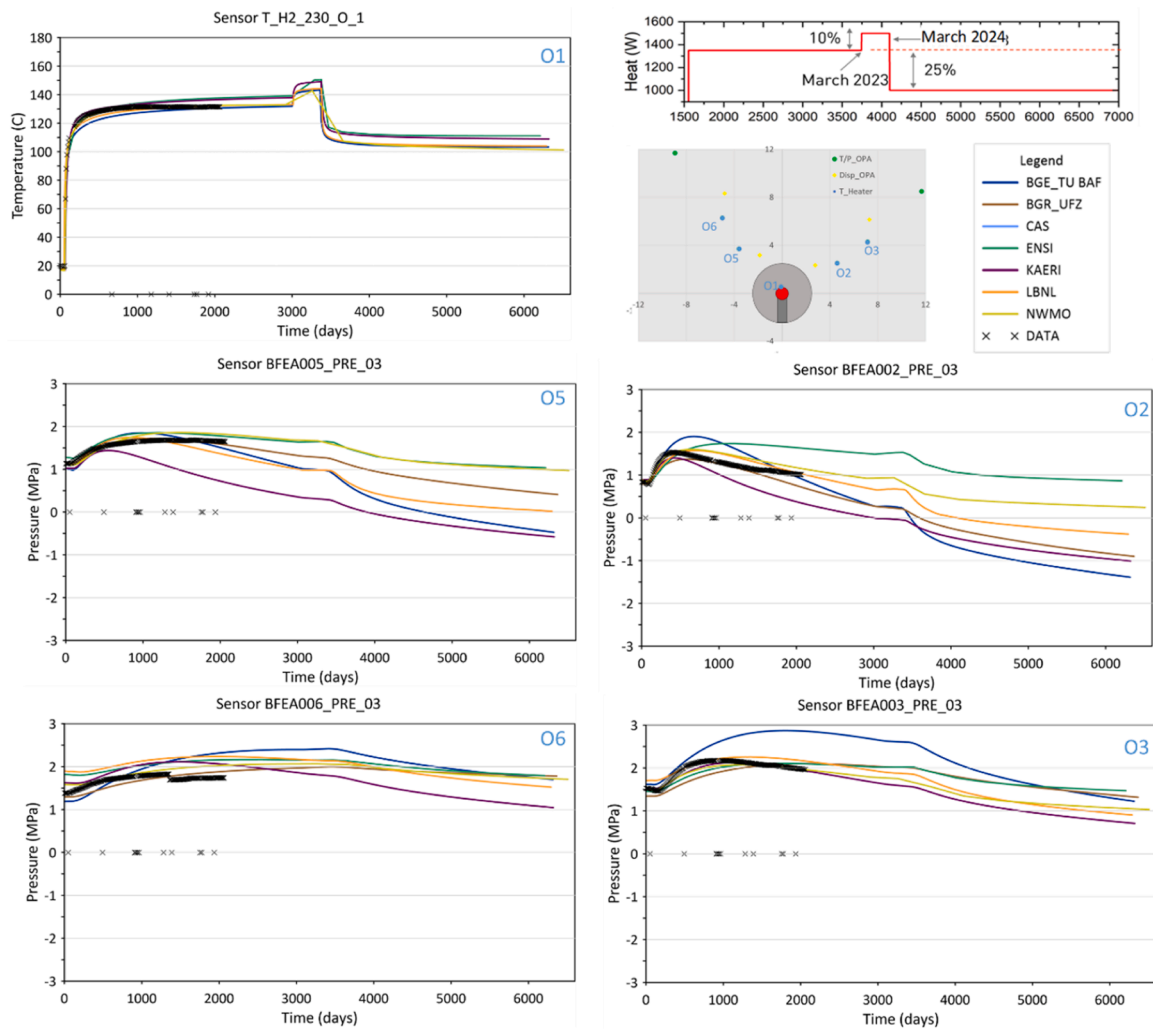


Fig. 13. Selected pressure results for locations in the Opalinus Clay from Step 3 and the temperature evolution at the heater (O1).

The work undertaken in this study highlights the importance of using a multi-team approach for making predictions about future evolution of repository systems. Firstly, there was a clear benefit in having several teams working on simplified models. It was not straightforward to reach agreement between the teams on the 2D benchmarks and the introduction of simplified cases for which analytic solutions were available was extremely helpful in resolving differences between the teams. Secondly, these are complicated models with many equations and parameters, so it is not uncommon for errors to arise. Having multiple teams against which results can be checked helps spot these errors and improve the predictions provided. Finally, moving from simple 2D models where good agreement has been achieved to 3D models with more freedom for modellers to make choices has introduced a spread in the predictions. This is one indication of the uncertainty involved in making such predictions and will constrain the range of potential future evolution of the repository system better than parameter variations alone.⁸ The difference in modelling approaches and calibration can be used to derive an envelope of performance and hence get an idea about uncertainty in both the models and the disposal system itself.

Conclusion

The teams were able to model the THM coupled processes relevant for the FE experiment and were successful in reproducing the data. The prediction of the development of temperature, pore pressure and relative humidity for the next 10 years after the modification of the heat

power provided an expectancy range for the measured parameters. Those results for the FE experiment were only achieved due to the rather extensive benchmark studies beforehand. This underlines the value of well-constrained benchmarking studies when comparing complex process model implementations.

Modelling the FE experiment has confirmed that the conceptual understanding of coupled THM processes documented by previous authors is valid. Our results indicate as well that damage induced in the rock by the excavation might have a significant impact on the heating-induced increase of pore pressure and its subsequent dissipation.

The properties and extent of damaged Opalinus Clay around tunnels have not been studied in detail in this work, but there is a clear need to consider this if pore pressure in the Opalinus Clay is to be well captured. Especially the pore pressures in the vicinity of the tunnel will be affected. With increasing distance, the influence will be less pronounced.

Despite this conceptual understanding being well documented in the literature and presented to teams in a modelling specification, it took significant effort for these experienced teams, using well-tested codes, to all achieve consistent results and there remains uncertainty as to why some team's results do not match. Thus, the schedule of work had to be shifted by several months. This illustrates the complexity of the task at hand. This is particularly useful for waste management organisations and their regulators to appreciate the value of using a group of modelling teams, codes and approaches when using models to predict the time evolution of their disposal system. The predictions of an ensemble of

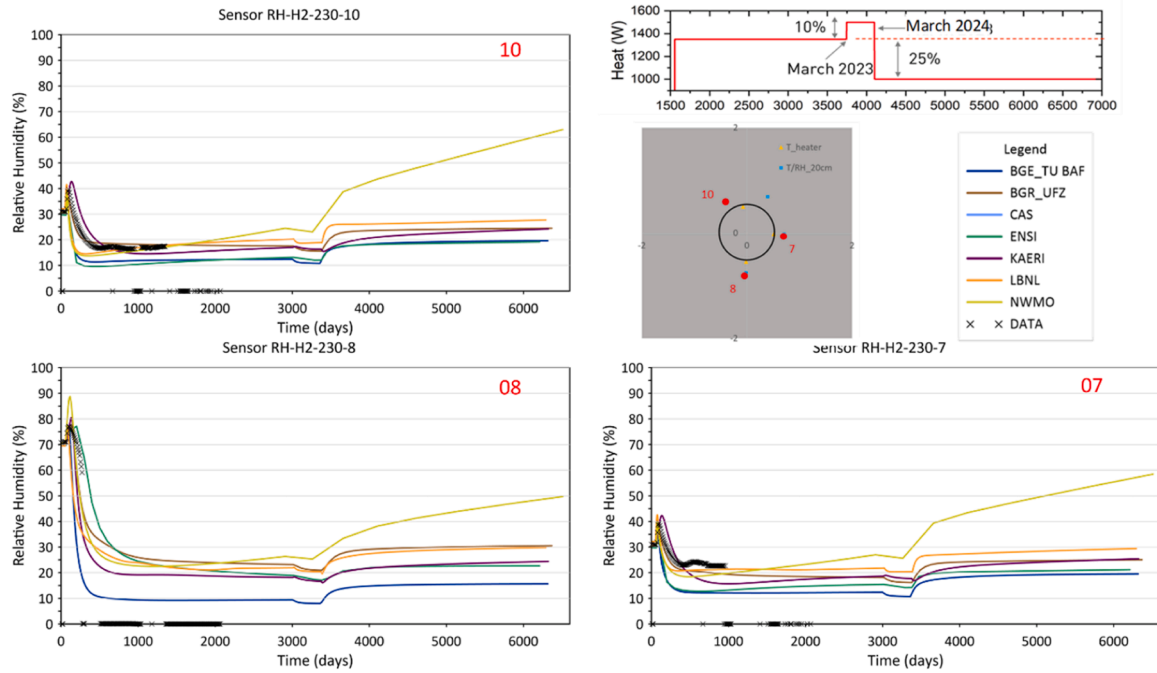


Fig. 14. Selected relative humidity results for locations in the tunnel from Step 3.

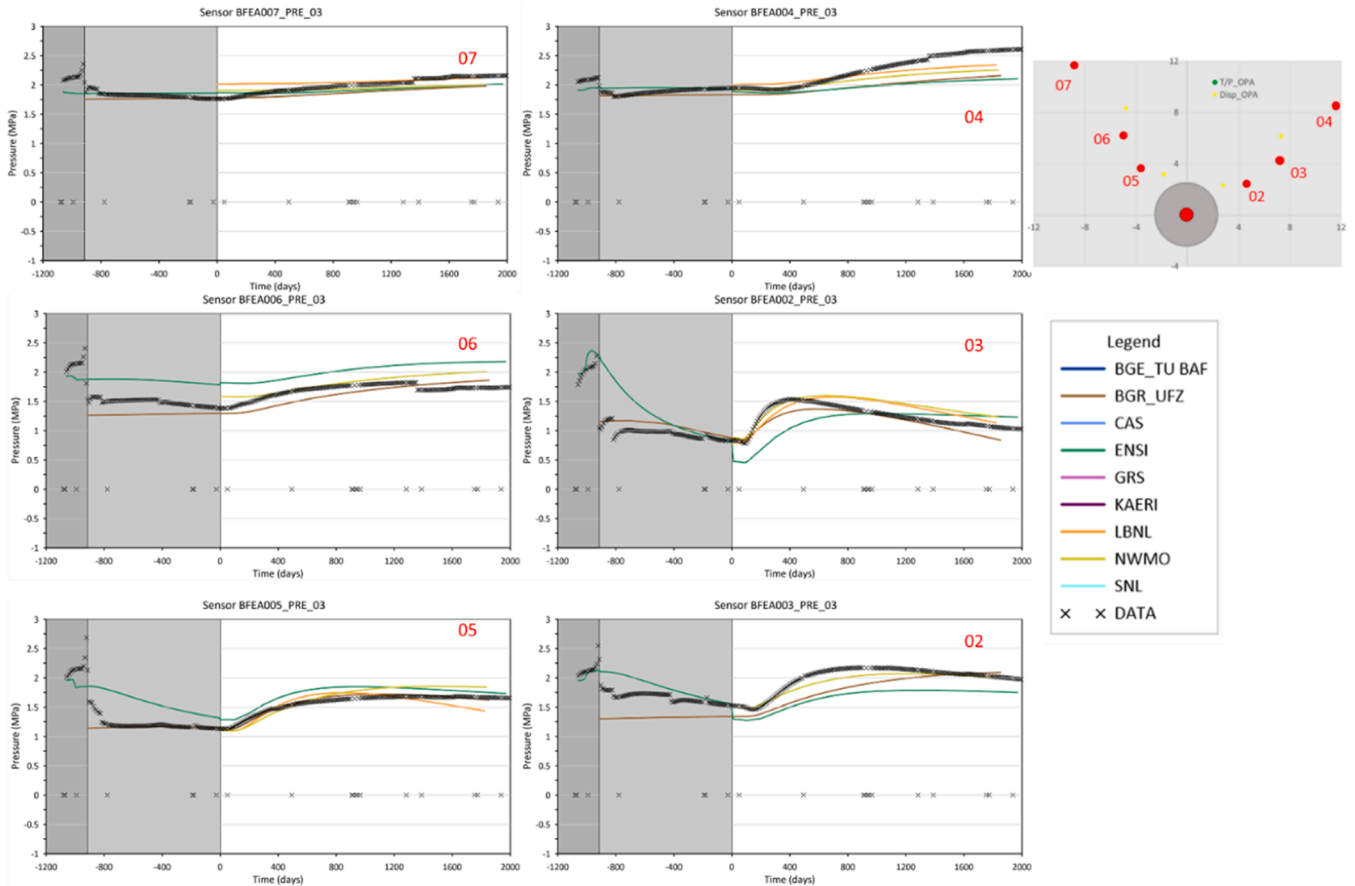


Fig. 15. Change in pore water pressure for the teams with a reduction of YM to 20 % (NWMO, LBNL, ENSI), 25 % (BGR/UFZ).

models will capture some of the uncertainties in the modelling process better than relying on a single modelling team.

CRedit authorship contribution statement

Edward Matteo: Writing – review & editing. **Bastian Johannes Graupner:** Writing – original draft. **Thomas Nagel:** Writing – review & editing. **Kate Thatcher:** Writing – original draft. **Rebecca Newson:** Writing – review & editing. **Larissa Friedenberg:** Writing – review & editing. **Peng-Zhi Pan:** Writing – review & editing. **Chan-Hee Park:** Writing – review & editing. **Changsoo Lee:** Writing – review & editing. **Taehyun Kim:** Writing – review & editing. **Ruiping Guo:** Writing – review & editing. **Michael Pitz:** Writing – review & editing. **Jonny Rutqvist:** Writing – review & editing. **Teklu Hadgu:** Writing – review & editing. **Jan Thiedau:** Writing – review & editing. **Wenbo Hou:** Writing – review & editing. **Luca Urpi:** Writing – review & editing. **Sonja Kaiser:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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